

FIRM: Fairness Interventions at Runtime and Model-training

Boyuan Chen, Arnav Nethula, Zachary Lee, Alex Kim,
Cole Blondin, Kevin Zhu, Sean O'Brien

Algoverse AI Research

boyuanchen@algoverseairesearch.org, akn74@scarletmail.rutgers.edu,
zn12@illinois.edu, alexjkim@stanford.edu,
cole@algoverseairesearch.org, kevin@algoverse.us, sean@algoverse.us

Abstract

We propose Fairness Interventions at Runtime and Model-training (FIRM), a unified framework for mitigating bias in modern NLP models. Such models often encode and perpetuate bias in opaque ways. Furthermore, existing mitigation approaches typically treat models as black boxes and apply static interventions that do not persist as the model evolves. Using paired counterfactual input, we apply causal tracking techniques to localize internal *features* (attention layer, attention heads and neurons) that are causally responsible for biased behavior. Once identified, we intervene through (1) pinpoint tuning that fine-tunes only the biased components during training, and (2) Debaised Steering Vectors (DSVs) that suppress biased representations during inference. FIRM achieves up to 47.6% mean bias reduction across multiple datasets.

1 Introduction

Modern NLP models are used in critical domains such as hiring, finance, healthcare, and law, yet often learn and perpetuate harmful social biases in opaque ways [Gallegos et al., 2024, Tan and Celis, 2019]. These biases can manifest as stereotyping or unfair treatment groups, causing real-world harm.

Yet current bias mitigation approaches face two key limitations. First, most fairness interventions are *static and lack longitudinal robustness* [Luo et al., 2023, Jin et al., 2020], being evaluated on a fixed snapshot without testing whether bias mitigation persists after fine-tuning or retraining [Ferrario and Loi, 2020].

Second, there remains a *disconnect between input-level counterfactual methods and internal model mechanisms*. Most treat the model as a black box: they generate altered inputs (e.g., using generative techniques like Polyjuice [Wu et al., 2021] or minimal edits like HotFlip [Ebrahimi et al., 2017]) and observe changes in outputs, without revealing which neurons, attention heads, or MLP layers drive bias. As a result, interventions lack transparency and may not generalize beyond the specific augmented data.

In this context, we introduce FIRM, an end-to-end bias mitigation system combining training-time and inference-time interventions to produce bias-reduced models. At training, we use Pinpoint Tuning [Chen et al., 2024] to localize and fine-tune the specific representations responsible for biased behavior. At inference time, we incorporate FairSteer [Li et al., 2025] style bias-vector steering. Our integrated pipeline targets both underlying representations that encode bias and the outputs where such bias manifests. Crucially, our proposed pipeline traces these bias-related features across multiple training cycles. After mitigating bias in the initial model, we retrain or fine-tune the model on new data, then reapply the same counterfactual probes to measure whether the activation patterns previously linked to biased behavior reappear, diminish, or shift to other layers or components.

Hence, our unified framework FIRM answers the following questions:

Layer alignment: Does aligning the inference-time intervention (steering) layer with the training-time diagnostic (pinpoint) layer lead to stronger debiasing outcomes? We hypothesize that the most effective debiasing interventions occur when the steering vector is applied at the same layer where pinpoint tuning reveals the maximal behavioral difference between biased and unbiased model trajectories. This is based on the idea that hidden representations encode tasks-relevant and bias-relevant signals at specific layers, which can be causally altered through localized interventions [Chandna et al., 2025].

Multi-layer robustness: What happens when steering is applied to downstream or unrelated layers? To test the robustness and locality of representational bias, we investigate the effect of steering downstream from the pinpointed layer as well as at nonoverlapping layers. We consider both isolated and joint steering strategies across multiple layers. This informs our understanding of how bias

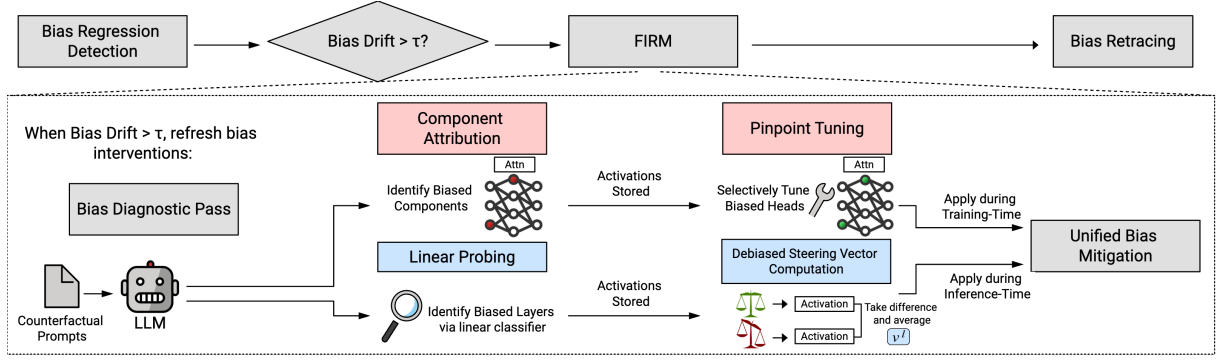


Figure 1: Overview of FIRM. Our pipeline contains 1) comprehensive dataset integration across 13 bias benchmarks covering gender, racial, religious, occupational, demographic, and sycophancy biases, 2) causal bias localization (training-time), fine-tuning of model components identified via interpretability techniques as responsible for biased outputs 3) dynamic bias steering (inference-time), real-time bias detection and correction using linear steering vectors applied during generation

propagates through the model and whether steering must be tightly localized.

Pipeline generalizability: How generalizable are our findings? We test on BBQ [Parrish et al., 2021], WinoBias [Zhao et al., 2018], CrowS-Pairs [Nangia et al., 2020], StereoSet [Nadeem et al., 2020], BOLD [Dhamala et al., 2021].

2 Methodology

We apply training-time pinpoint tuning with inference time steering, with a drift monitor triggering reanalysis when a threshold of representational drift exceeds τ .

2.1 Bias Diagnostic Pass: The Bias Diagnostic Pass is a single forward pass that stores activations, and serves as both a bias attribution and a bias detection mechanism. Attention layers and attention head activations from the forward pass are stored in a component registry, which is read by both selective debiasing training and steering vector computation.

2.2 Pinpoint Tuning: Using counterfactual bias datasets including demographic fairness and stereotype reduction, we selectively tune the biased attention heads identified in the component registry. We obtain a debiased model with selective parameter updates targeting only bias-causing components.

2.3 Debiased Steering Vector (DSV) Computation: The DSV uses Bias Diagnostic results to compute steering vectors using contrastive prompt-pairs. DSV is derived by subtracting residual stream representations of biased vs. counterfactual prompts at a specific diagnostic layer. During inference, the DSV is subtracted from the residual stream to suppress biased activations without

altering the model’s parameters.

2.4 Pipeline Setup and Evaluation: combining Pinpoint Tuning(training-time) and DSV (inference-time), we evaluate the pipeline on comprehensive bias benchmarks (BBQ, WinoBias, CrowS-Pairs, StereoSet, BOLD). The pipeline is evaluated on bias reduction, fairness, demographic parity. Importantly, the datasets used for bias localization and intervention are *disjoint* from the evaluation datasets.

2.5 Tracing Bias Flow with Patching: We then evaluate the pipeline (1) training-only (Pinpoint Tuning), (2) inference-only (DSV), and (3) joint application. We run activation patching and causal tracing to visualize where bias signals emerge and how they shift each intervention.

3 Results

Preliminary experiments indicate that applying the inference-time Debiased Steering Vector (DSV) at the same layer identified during training-time Pinpoint Tuning yields substantially lower bias scores across multiple benchmarks. FIRM achieves up to 47.6% mean bias reduction while preserving task accuracy. Bias mitigation appears strongest when steering vectors target the diagnostic layers revealed during training (Optimal layers 14–16), with reductions of up to 65% for inference-only and 58.9% for training-only methods. Ongoing path patching and causal tracing aim to verify bias-prone conduits and confirm that the relevant layers align with layers 14–16. Further work will assess the persistence of mitigation under model drift. All codes are publicly available at github.com/Zachbot168/Algoverse.

References

- Bhavik Chandna, Zubair Bashir, and Procheta Sen. 2025. [Dissecting bias in llms: A mechanistic interpretability perspective](#). *ArXiv*, abs/2506.05166.
- Wei Chen, Zhen Huang, Liang Xie, Binbin Lin, Houqiang Li, Le Lu, Xinmei Tian, Deng Cai, Yonggang Zhang, Wenxiao Wang, Xu Shen, and Jieping Ye. 2024. [From yes-men to truth-tellers: Addressing sycophancy in large language models with pinpoint tuning](#). *ArXiv*, abs/2409.01658.
- J. Dhamala, Tony Sun, Varun Kumar, Satyapriya Krishna, Yada Pruksachatkun, Kai-Wei Chang, and Rahul Gupta. 2021. [Bold: Dataset and metrics for measuring biases in open-ended language generation](#). *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*.
- Javid Ebrahimi, Anyi Rao, Daniel Lowd, and Dejing Dou. 2017. Hotflip: White-box adversarial examples for text classification. *arXiv preprint arXiv:1712.06751*.
- Andrea Ferrario and Michele Loi. 2020. A series of unfortunate counterfactual events: The role of time in counterfactual explanations. *arXiv preprint arXiv:2010.04687*.
- Isabel O Gallegos, Ryan A Rossi, Joe Barrow, Md Mehrab Tanjim, Sungchul Kim, Franck Dernoncourt, Tong Yu, Ruiyi Zhang, and Nesreen K Ahmed. 2024. Bias and fairness in large language models: A survey. *Computational Linguistics*, 50(3):1097–1179.
- Xisen Jin, Francesco Barbieri, Brendan Kennedy, Aida Mostafazadeh Davani, Leonardo Neves, and Xiang Ren. 2020. On transferability of bias mitigation effects in language model fine-tuning. *arXiv preprint arXiv:2010.12864*.
- Yichen Li, Zhiting Fan, Ruizhe Chen, Xiaotang Gai, Luqi Gong, Yan Zhang, and Zuozhu Liu. 2025. [Fairsteer: Inference time debiasing for llms with dynamic activation steering](#). *ArXiv*, abs/2504.14492.
- Yun Luo, Zhen Yang, Fandong Meng, Yafu Li, Jie Zhou, and Yue Zhang. 2023. An empirical study of catastrophic forgetting in large language models during continual fine-tuning. *arXiv preprint arXiv:2308.08747*.
- Moin Nadeem, Anna Bethke, and Siva Reddy. 2020. [Stereoset: Measuring stereotypical bias in pretrained language models](#). In *Annual Meeting of the Association for Computational Linguistics*.
- Nikita Nangia, Clara Vania, Rasika Bhalerao, and Samuel R. Bowman. 2020. [Crows-pairs: A challenge dataset for measuring social biases in masked language models](#). In *Conference on Empirical Methods in Natural Language Processing*.
- Alicia Parrish, Angelica Chen, Nikita Nangia, Vishakh Padmakumar, Jason Phang, Jana Thompson, Phu Mon Htut, and Sam Bowman. 2021. [Bbq: A hand-built bias benchmark for question answering](#). In *Findings*.
- Yi Chern Tan and L Elisa Celis. 2019. Assessing social and intersectional biases in contextualized word representations. *Advances in neural information processing systems*, 32.
- Tongshuang Wu, Marco Tulio Ribeiro, Jeffrey Heer, and Daniel S Weld. 2021. Polyjuice: Generating counterfactuals for explaining, evaluating, and improving models. *arXiv preprint arXiv:2101.00288*.
- Jieyu Zhao, Tianlu Wang, Mark Yatskar, Vicente Ordonez, and Kai-Wei Chang. 2018. [Gender bias in coreference resolution: Evaluation and debiasing methods](#). *ArXiv*, abs/1804.06876.